

Chemical bonding & hybridisation — Issue #33 candidate:

Source questions

Q1 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical bonding and hybridisation

Concept: Electron-domain geometry versus molecular shape

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Domain count before hybrid label

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

BF₃ and NH₃ each have three sigma bonds to the central atom. BF₃ has no central lone pair; NH₃ has one. Explain why three

Representation

sigma bonds alone cannot determine the same central hybridisation and molecular shape for both.

Conditions

- Use the bounded introductory localized hybrid-orbital model requested by the question; do not treat molecular shape alone as direct measurement of orbital mixing.

Common wrong route • open if you want a warning

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Supplied three-bond comparison



Connectivity layout only; page angles are not measured ge
Supplied situation

Need a hint? • guided support

+

Need the concept again?

Electron-domain geometry versus molecular shape

Q2 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical bonding and hybridisation

Concept: Sigma/pi orbital inventory without double-counting

Concept difficulty: Hard concept

Question difficulty: Connected reasoning

Family: Sigma/pi orbital allocation

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

For $\text{H}_2\text{C}=\text{O}$, assign the carbon hybridisation, count the sigma and pi components of the $\text{C}=\text{O}$ bond, and identify the carbon orbital left available for the pi component. Keep electron-domain counting separate from counting shared electron pairs.

Representation

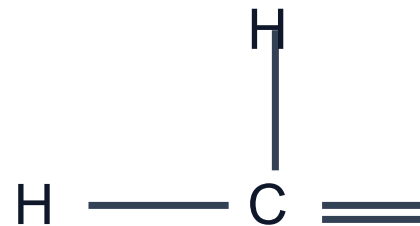
Conditions

- Use the bounded introductory localized hybrid-orbital model requested by the question; do not treat molecular shape alone as direct measurement of orbital mixing.

Common wrong route · open if you want a warning

+

H₂C=O connectivity



H₂C=O connectivity

Connectivity layout only; page angles are not measured

Supplied situation

Need a hint? · guided support

+

Need the concept again?

Sigma/pi orbital inventory without double-counting

Q3 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical bonding and hybridisation

Concept: Sigma/pi orbital inventory without double-counting

Concept difficulty: Hard concept

Question difficulty: Connected reasoning

Family: Sigma/pi orbital allocation

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

Ethene is $\text{H}_2\text{C}=\text{CH}_2$. Translate this Lewis representation into a qualitative orbital-overlap diagram showing the C–C sigma framework, the two unhybridised carbon p orbitals and their side-on overlap. Label which directions are in the molecular plane and which are perpendicular to it.

Representation

Conditions

- Use the bounded introductory localized hybrid-orbital model requested by the question; do not treat molecular shape alone as direct measurement of orbital mixing.

Common wrong route · open if you want a warning

+

Ethene Lewis connectivity



Ethene connectivity

Connectivity layout only; page angles are not measured

Supplied situation

Need a hint? · guided support

+

Need the concept again?

Sigma/pi orbital inventory without double-counting

COMPETITION TRANSFER

Close an orbital inventory under a changed local description

Original transfer task; not claimed as an official exam/PYQ item.

A neutral carbon centre is described by three coplanar sigma-bond directions and one perpendicular p-orbital participation. Without using a memorised molecule name, give an orbital-job inventory and state one check that would expose double-counting.

Q4 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry **Topic:** Chemical bonding and hybridisation

Concept: Sigma/pi orbital inventory without double-counting **Concept difficulty:** Hard concept

Question difficulty: Connected reasoning **Family:** Sigma/pi orbital allocation **Question type:** Constructed response

Source: Owner-supplied question **Provenance:** Owner-supplied question

ATTEMPT FIRST

For CO₂, connect the central-carbon Lewis domain count to its hybrid-orbital model, the number of remaining carbon p orbitals and the sigma/pi composition of its two C=O bonds. Explain how these pieces fit one consistent linear arrangement.

Conditions

- Use the bounded introductory localized hybrid-orbital model requested by the question; do not treat molecular shape alone as direct measurement of orbital mixing.

Common wrong route · open if you want a warning

+

Representation

CO₂ Lewis connectivity



CO₂ connectivity

Connectivity layout only; page angles are not measured

Supplied situation

Need a hint? · guided support

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Need the concept again?

Sigma/pi orbital inventory without double-counting

Q5 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical bonding and hybridisation

Concept: Domain count before hybridisation label

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Domain count before hybrid label

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

Consider the claim: 'Any central atom with three sigma bonds must be sp^2 .' Given BF_3 : three B–F sigma bonds, zero boron lone pairs, planar; NH_3 : three N–H sigma bonds, one nitrogen lone pair, pyramidal. Decide whether these observations

Representation

support the claim, give the decisive warrant, and state a more defensible introductory rule. Explain what the shape observations alone do not prove about orbitals.

Conditions

- Use the bounded introductory localized hybrid-orbital model requested by the question; do not treat molecular shape alone as direct measurement of orbital mixing.

Common wrong route • open if you want a warning

+

Supplied BF₃ and NH₃ counterexample candidate



BF₃ Lewis inventory

NH₃ Lewis

Connectivity layout only; page angles are not measured geometrically
Supplied situation

Need a hint? • guided support

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Need the concept again?

Domain count before hybridisation label

Q6 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical bonding and hybridisation

Concept: Domain count before hybridisation label

Concept difficulty: Medium concept

Question difficulty: Direct application

Family: Domain count before hybrid label

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

In CH₃F, carbon has three C–H single bonds and one C–F single bond. Assign its introductory local hybridisation and explain why the number of electron-domain directions is unchanged from CH₄ even though the attached atoms are different. Do not assume all bond properties are identical.

Representation

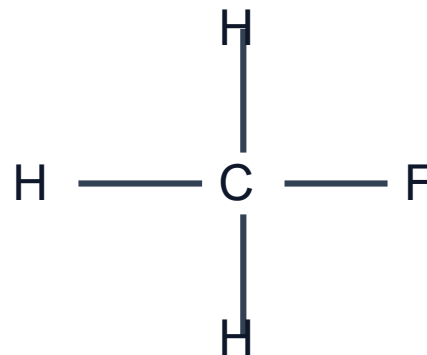
Conditions

- Use the bounded introductory localized hybrid-orbital model requested by the question; do not treat molecular shape alone as direct measurement of orbital mixing.

Common wrong route · open if you want a warning

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CH₃F and CH₄ connectivity



CH₃F connectivity



CH₄ con

Connectivity layout only; page angles are not measured ge
Supplied situation

Need a hint? · guided support

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Need the concept again?

Domain count before hybridisation label

Q7 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical bonding and hybridisation

Concept: Sigma/pi orbital inventory without double-counting

Concept difficulty: Hard concept

Question difficulty: Connected reasoning

Family: Sigma/pi orbital allocation

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

HCN is represented as $\text{H}-\text{C}\equiv\text{N}$ with a lone pair on nitrogen. Assign the introductory hybridisation at carbon and nitrogen, and count the sigma and pi components of the $\text{C}\equiv\text{N}$ bond. Explain how the local domain count is compatible with those components.

Representation

Conditions

- Use the bounded introductory localized hybrid-orbital model requested by the question; do not treat molecular shape alone as direct measurement of orbital mixing.

Common wrong route · open if you want a warning

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Supplied HCN Lewis connectivity



HCN connectivity; : is the 1

Connectivity layout only; page angles are not meas

Supplied situation

Need a hint? · guided support

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Need the concept again?

Sigma/pi orbital inventory without double-counting

Q8 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical bonding and hybridisation

Concept: Electron-domain geometry versus molecular shape

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Electron-domain geometry versus molecular shape

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

For methylamine, $\text{CH}_3\text{-NH}_2$, carbon has four single bonds and nitrogen has three single bonds plus a lone pair. Compare the introductory local hybridisation and arrangement of bonded atoms at carbon and nitrogen. Explain why the same

Representation

hybridisation label does not require identical molecular shapes.

Conditions

- Use the bounded introductory localized hybrid-orbital model requested by the question; do not treat molecular shape alone as direct measurement of orbital mixing.

Common wrong route · open if you want a warning

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Supplied methylamine connectivity



Nitrogen lone pair is n

Connectivity layout only; page angles are not measured

Supplied situation

Need a hint? · guided support

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Need the concept again?

Electron-domain geometry versus molecular shape

Q9 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical bonding and hybridisation

Concept: Sigma/pi orbital inventory without double-counting

Concept difficulty: Hard concept

Question difficulty: Connected reasoning

Family: Sigma/pi orbital allocation

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

A carbon centre has three sigma-bond directions, no lone pair, and participates in one pi bond. Translate this description into a qualitative local orbital arrangement and check that the orbitals used in the sigma and pi parts have not been counted twice.

Representation

Conditions

- Use the bounded introductory localized hybrid-orbital model requested by the question; do not treat molecular shape alone as direct measurement of orbital mixing.

Common wrong route · open if you want a warning

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Supplied local carbon jobs

Three sigma directions; no lone pair

One pi participation

Available valence basis: one 2s and three

Connectivity layout only; page angles are not measured

Supplied situation

Need a hint? · guided support

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Need the concept again?

Sigma/pi orbital inventory without double-counting

COMPETITION TRANSFER

Close an orbital inventory under a changed local description

Original transfer task; not claimed as an official exam/PYQ item.

A neutral carbon centre is described by three coplanar sigma-bond directions and one perpendicular p-orbital participation. Without using a memorised molecule name, give an orbital-job inventory and state one check that would expose double-counting.

Q10 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical bonding and hybridisation

Concept: Electron-domain geometry versus molecular shape

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Electron-domain geometry versus molecular shape

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

A student says, 'If the electron-domain geometry is tetrahedral, the atoms must form a tetrahedral molecular shape.'

Evaluate the statement using CH₄, NH₃ and H₂O, with their respective central lone-pair counts of zero, one and two.

Conditions

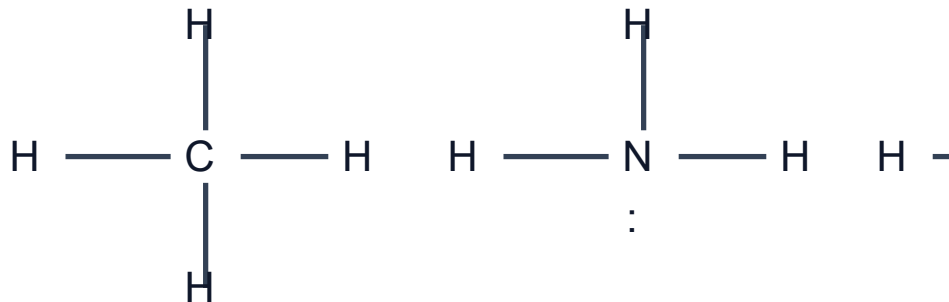
- Use the bounded introductory localized hybrid-orbital model requested by the question; do not treat molecular shape alone as direct measurement of orbital mixing.

Common wrong route · open if you want a warning

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Representation

Supplied four-domain comparisons



CH₄ connectivity

NH₃ Lewis inventory

H₂O I

Connectivity layout only; page angles are not measured geometry.

Supplied situation

Need a hint? · guided support

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Need the concept again?

Electron-domain geometry versus molecular shape

COMPETITION TRANSFER

Hold domain geometry fixed while changing atom occupancy

Original transfer task; not claimed as an official exam/PYQ item.

A four-domain centre is changed from four bonds to two bonds plus two lone pairs without changing the total domain count. Explain which geometry description is preserved and which atom-only description must be reconsidered.